

Cuong Manh Nguyen

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Sex: Male | Date of birth: 05 Dec 1999 | Nationality: Vietnamese

I am a dedicated Computer Science graduate with a solid foundation in artificial intelligence. My passion for this area is driven by the rapid advancements in technology and its remarkable potential to improve everyday life. I aim to deepen my expertise so that I can contribute to groundbreaking AI projects that push the boundaries of the field and enrich our daily interactions with technology.

EDUCATION

Bachelor's in Information Technology

Post & Telecommunication Institute of Technology (PTIT)
GPA: 3.16/4.0 (7.93/10)

Ha Noi, Vietnam
August 2017 - Dec 2021

Bachelor thesis: Research to improve the quality of recommendation systems (Score: 10/10 - Top 0.01%)

PUBLICATIONS

Cuong Manh Nguyen, Huy-Hoang Do-Huu, Viet Thanh Duy Nguyen, Truong-Son Hy. "TeBaAb: Text-based antigen-conditioned antibody redesign." Submitted to the NeurIPS 2025 Workshop, with the aim of publishing in the Oxford Journal.

Cuong Manh Nguyen, Xuan Bach Nguyen, Minh Phuong Tu. "Combining User Specific and Global News Features for Neural News Recommendation." ACIIDS 2022 (Springer LNCS, CORE B/Q2, Info Systems/AI).

Cuong Manh Nguyen, et al. "Learning Vietnamese-English Code-Switching Speech Synthesis Model Under Limited Code-Switched Data Scenario" PRICAI 2021 (Springer LNAI, CORE B/Q2, AI/ML).

Thi Cuc Bui, **Cuong Manh Nguyen**, et al. "Multi-Task Text Normalization Approach for Speech Synthesis." PRICAI 2021 (Springer LNAI, CORE B/Q2, AI/ML).

Cuong Manh Nguyen, et al. "Development of Smartcall Vietnamese Text-to-Speech for VLSP 2020." VLSP 2020 (ACL, leading regional workshop on Vietnamese NLP).

WORK EXPERIENCE

HySonLab, University of Alabama, USA

HySonLab focuses on AI for Science, Bioinformatics, Drug Discovery, Drug Repurposing, Natural Language Processing, and Computer Vision.

Remote

April 2025 - Present

Research Intern

Conducted research on drug discovery and repurposing, with a focus on protein and antibody generation.

Zalo AI, VNG Corporation

Zalo AI is the artificial intelligence research and development center of VNG - Vietnam's first technology unicorn. Its goal is to apply AI technology to improve and enhance the daily lives of Vietnamese people.

Ho Chi Minh City, Vietnam

Feb 2020 - Present

AI Engineer

- **Product KiLM** - a large language model developed by Zalo AI, specifically optimized for Vietnamese. In benchmark tests, its KiLM-13b-v24.7.1 version scored 66.07, closely trailing Llama-3-70B at 66.44.
 - Contributed to an AI-based creative card product by refactoring service from Python to Golang, increasing concurrent user response by 500%.
 - Developed a Retrieval-Augmented Generation system for KiLM, integrating Milvus for vector search, Elasticsearch for FAQ retrieval, and MinIO for scalable storage, enhancing information retrieval accuracy and response efficiency.

- **Product Kiki Auto** - an intelligent Vietnamese voice assistant integrated into cars, enabling hands-free navigation, music, and information access in cars.
 - Conducted in-depth research on Vietnamese text-to-speech (TTS) technologies, experimenting with state-of-the-art models such as Tacotron2, FastSpeech, and VITS.
 - Successfully refactored TTS service, resulting in an 80% error rate reduction and significantly improved performance and reliability.
 - Expanded and improved voice processing systems (ASR services). Enhanced system resilience and scalability with advanced fault tolerance measures.

EXTRACURRICULAR ACTIVITY

Programming PTIT Club, Post & Telecommunication Institute of Technology **Ha Noi, Vietnam**
President *Dec 2019 - Dec 2020*

- Organized fundamental programming lectures and AI courses for students.
- Organized academic competitions and events that encouraged collaboration among club members.
- Managed club activities related to academics, communications, finance, and human resources.

Mid-Autumn for Children - Lighting Up Vietnamese Childhood **Ha Noi, Vietnam**
Group leader *Aug 2017 - Nov 2017*

- Directly advise and mobilize people to participate in voluntary blood donation.
- Managed team members and supported receiving blood donation applications on the official blood donation day.

AWARDS

- Encourage Academic Scholarship PTIT *2018, 2019, 2020*
- Top 10 Finalists - National Digital Race *2020*
- First Prize in the Text-to-Speech category at the Vietnamese Language and Speech Processing Conference *2020*
- Third Prize at the ACM/ICPC PTIT *2019*
- Consolation Prize at the ACM/ICPC PTIT *2018*

TECHNICAL SKILLS

- Excellent use of Python for developing machine-learning models.
- Highly skilled in developing artificial intelligence services using Python and Golang.
- Possesses experience working with various machine learning frameworks and toolkits, including TensorFlow, Keras, and PyTorch.
- **Technologies:** Linux, Git, Kubernetes, Jira, CUDA, FastAPI, MinIO, Redis, Elasticsearch, MongoDB, Grafana, Prometheus, Milvus.

REFEREES

- **Prof. Truong-Son Hy**
 Department of Computer Science at the University of Alabama at Birmingham, USA
 Email: thy@uab.edu
- **Prof. Phuong Minh Tu**
 Faculty of Information Technology, Posts and Telecommunications Institute of Technology, Vietnam
 Email: phuongtm@ptit.edu.vn
- **Dr. Dat Ba Nguyen**
 Senior Product Manager of Zalo AI - VNG Corporation, Vietnam
 Email: datnb2@vng.com.vn